

Team Names:

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```
// Test addStudent and getLastStudentAdded
System.out.println("Testing addStudent...");
ms.addStudent(studentId:1);
System.out.println("Last student added: " + ms.getLastStudentAdded() + " (Expected: 1)");

// Test addCourse and getLastCourseAdded
System.out.println("\nTesting addCourse...");
ms.addCourse(courseId:101);
System.out.println("Last course added: " + ms.getLastCourseAdded() + " (Expected: 101)");

// Test enrollStudent

// Test enrollStudent
System.out.println("\nTesting enrollStudent...");
ms.enrollStudent(studentId:1, courseId:101);
System.out.println("Courses for student 1:");
ms.listCoursesByStudent(studentId:1);
System.out.println("Students in course 101:");
ms.listStudentsByCourse(courseId:101);
```

```
/557d4671646b6b1f5b17c5b2646b2426/TestAdd.java
Testing addStudent...
Last student added: 1 (Expected: 1)

Testing addCourse...
Last course added: 101 (Expected: 101)

Testing enrollStudent...
Courses for student 1:
Course Id: 101, Title: null

Students in course 101:
Student Id: 1, Name: null
```

```
// Test sortStudentsById
System.out.println("\nTesting sortStudentsById...");
ms.addStudent(studentId:3);
ms.addStudent(studentId:2);
ms.enrollStudent(studentId:2, courseId:101);
ms.enrollStudent(studentId:3, courseId:101);

System.out.println("Students in course 101 before sorting:");
ms.listStudentsByCourse(courseId:101);
ms.sortStudentsById(courseId:101);
System.out.println("Students in course 101 after sorting:");
ms.listStudentsByCourse(courseId:101);
```

Students in course 101 after removal:

```
Testing sortStudentsById...
Students in course 101 before sorting:
Student Id: 3, Name: null
Student Id: 2, Name: null

Students in course 101 after sorting:
Student Id: 2, Name: null
Student Id: 3, Name: null

Testing sortCoursesById...
Courses for student 1 before sorting:
Course Id: 102, Title: null
Course Id: 100, Title: null

Courses for student 1 after sorting:
Course Id: 100, Title: null
Course Id: 102, Title: null
```

```
// Test isFullCourse
System.out.println("\nTesting isFullCourse...");
System.out.println("Is course 101 full? " + ms.isFullCourse(courseId:101) + " (Expected: false)");
// Enroll 30 students in course 101
for (int i = 3; i <= 30; i++) {
    ms.addStudent(i);
    ms.enrollStudent(i, courseId:101);
}
System.out
    .println("Is course 101 full after adding 30 students? " + ms.isFullCourse(courseId:101) + " (Expected: true)");

// Test isNormalStudent
System.out.println("\nTesting isNormalStudent...");
System.out.println("Is student 1 normal? " + ms.isNormalStudent(studentId:1) + " (Expected: true)");
// Enroll student 1 in more courses
for (int i = 103; i <= 108; i++) {
    ms.addCourse(i);
    ms.enrollStudent(studentId:1, i);
}
System.out.println(
    "Is student 1 normal after enrolling in 9 courses? " + ms.isNormalStudent(studentId:1) + " (Expected: false)");
```

```
// Test undo
System.out.println("\nTesting undo...");
ms.undo(); // Undo last enrollment (course 108)

System.out.println("After undo, is student 1 normal? " + ms.isNormalStudent(studentId:1) + " (Expected: true)");

// Test redo
System.out.println("\nTesting redo...");
ms.redo(); // Redo the enrollment

System.out.println("After redo, is student 1 normal? " + ms.isNormalStudent(studentId:1) + " (Expected: false)");

// Test removeStudent
System.out.println("\nTesting removeStudent...");
ms.removeStudent(studentId:1);
System.out.println("Courses for student 1 after removal:");
try {
    ms.listCoursesByStudent(studentId:1);
} catch (Exception e) {
    System.out.println("Student not found (Expected)");
}
```

```
Testing isFullCourse...
Is course 101 full? false (Expected: false)
Is course 101 full after adding 30 students? true (Expected: true)

Testing isNormalStudent...
2
Is student 1 normal? true (Expected: true)
8
Is student 1 normal after enrolling in 9 courses? false (Expected: false)

Testing undo...
7
After undo, is student 1 normal? true (Expected: true)

Testing redo...
8
After redo, is student 1 normal? false (Expected: false)

Testing removeStudent...
Courses for student 1 after removal:
Student not found
```

```
// Test removeCourse
System.out.println("\nTesting removeCourse...");
ms.removeCourse(courseId:101);

System.out.println("Students in course 101 after removal:");
try {
    ms.listStudentsByCourse(courseId:101);
} catch (Exception e) {
    System.out.println("Course not found (Expected)");
}
```

```
Testing removeCourse...
Students in course 101 after removal:
Course not found
```